

Comparison Report

Leveraging Machine Learning Models for Seismic Prediction: A Study on Model Performance and Optimization.

The main parameters used for comparison are:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R² Score
- Accuracy

Dataset and Model Comparison

The reference paper uses the Indian Subcontinent Earthquake dataset compiled from the USGS Earthquake Catalogue, covering November 2000 to November 2024. The paper states that the dataset contains 17,805 samples and 22 features.

The updated notebook uses the same general earthquake-prediction dataset and implements the same six principal models used in the reference paper.

Therefore, the model-level comparison is directly relevant.

Performance Comparison

Model	Paper MAE	Current MAE	Paper RMSE	Current RMSE	Paper R ²	Current R ²	Paper Accuracy	Current Accuracy
XGBoost	0.3235	0.259053	0.4387	0.366845	0.1415	0.270383	92.76%	95.60%
Random Forest	0.3333	0.269857	0.4501	0.380107	0.0966	0.216678	92.54%	95.42%
Stacking Regressor	0.3228	0.269916	0.4369	0.382277	0.1488	0.207707	92.78%	95.42%
LSTM	0.3420	0.346189	0.4559	0.433292	0.0730	- 0.017863	92.35%	94.12%
KNN	0.3379	0.306288	0.4571	0.427643	0.0681	0.008502	92.44%	94.80%

Model	Paper MAE	Current MAE	Paper RMSE	Current RMSE	Paper R ²	Current R ²	Paper Accuracy	Current Accuracy
SVM	0.3759	0.280251	0.5278	0.421845	-0.2422	0.035206	91.59%	95.42%

Final Conclusion

The comparison shows that the current implementation generally produces better results than the reference paper for most of the matched models.

The current **XGBoost model is the best-performing model**, achieving:

- MAE = **0.259053**
- RMSE = **0.366845**
- R² = **0.270383**
- Accuracy = **95.60%**

Compared with the reference paper, the current implementation improves the performance of XGBoost, Random Forest, Stacking Regressor, KNN and SVM across several important error and goodness-of-fit measures. LSTM shows mixed results, with a lower RMSE but a negative R².

The overall model ranking also supports the conclusion of the reference paper that **ensemble-based methods, particularly XGBoost, are suitable for earthquake magnitude prediction**. The reference paper concludes that XGBoost achieved the best performance after hyperparameter adjustment.